Exp No: 9	MODEL DEPLOYMENT: REST API WITH FLASK AND CONTAINERIZATION WITH DOCKER

AIM:

To deploy a trained machine learning model as a web service using a Flask REST API and package the application with Docker for portable and consistent execution.

ALGORITHM:

1. Prepare a machine learning model and save it to a file.
2. Create a Flask application and load the saved model.
3. Define a POST endpoint that accepts JSON input.
4. Validate the input, generate a prediction, and return the result as JSON.
5. Create a requirements file and Dockerfile.
6. Build a Docker image and run the application in a container.
7. Test the API and verify the prediction response.

CODE:

```
from flask import Flask, request, jsonify
import pickle
import numpy as np
from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier

# Train and save a sample model if it does not already exist
try:
    with open("model.pkl", "rb") as file:
        model = pickle.load(file)
except FileNotFoundError:
    iris = load_iris()
    model = DecisionTreeClassifier(random_state=42)
    model.fit(iris.data, iris.target)
    with open("model.pkl", "wb") as file:
        pickle.dump(model, file)

app = Flask(__name__)

@app.route("/", methods=["GET"])
def home():
    return jsonify({"message": "Model Deployment API is running"})

@app.route("/predict", methods=["POST"])
def predict():
    data = request.get_json(silent=True)

    if not data or "features" not in data:
        return jsonify({"error": "Please provide features as a JSON list"}), 400

    try:
        features = np.array(data["features"], dtype=float)
        if features.ndim != 1 or len(features) != 4:
            return jsonify({"error": "Exactly 4 numeric features are required"}), 400

        prediction = model.predict(features.reshape(1, -1))
        return jsonify({"prediction": str(prediction[0])})
    except (ValueError, TypeError) as error:
        return jsonify({"error": str(error)}), 400

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

OUTPUT:

Test the home endpoint:

```
GET http://localhost:5000/
{
  "message": "Model Deployment API is running"
}
```

```
Request:
{
  "features": [5.1, 3.5, 1.4, 0.2]
}

Example response:
{
  "prediction": "0"
}
```

RESULT:

The machine learning model was deployed using a Flask REST API. The API accepts input data in JSON format and returns predictions as JSON responses. Docker packages the application and its dependencies into a container, enabling consistent execution across supported environments.